Trade Hubs and the Uneven Gains from Globalization: Urban and Rural Disparities from Satellite Nighttime Lights Data

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Abstract

This study examines how export growth affects local economic activity and rural–urban inequality, using nighttime luminosity as a proxy for income. We construct bilateral export values for each country pair from 1997 to 2020, covering countries with at least one major port. Trade-hub areas are defined as locations within a 50 km radius (30 km in robustness checks) of each country's three largest ports and three largest international airports. To account for zero export values, we estimate a gravity equation using Poisson pseudo–maximum likelihood. To address export endogeneity, we employ a two-stage least squares (2SLS) framework in which export volumes are instrumented using predicted trade from a dynamic gravity equation that captures exogenous variation arising from changes in transportation costs across modes. To avoid mechanically capturing lights from ports and airports themselves, we exclude nighttime luminosity within a 3 km radius of each port and airport.

Our results show that a 1 percent increase in exports raises nighttime luminosity by 0.31 percent in trade-hub areas and by 0.06 percent in non-trade-hub areas, widening the rural–urban income gap by an estimated 15–17 percent between 1997 and 2018. Export growth also increases nighttime luminosity per capita by 0.19 percent in trade-hub areas and by 0.04 percent elsewhere, although modest migration into trade hubs partially offsets per-capita gains. Overall, the results indicate that export growth has heterogeneous spatial effects, with substantially larger gains in trade-hub areas, thereby contributing to widening rural–urban income disparities.

1 Introduction

Economic activity is unevenly distributed within countries. Nightlight data reveal that brighter areas—indicative of higher economic activity—are often located near coastlines, rivers, or canals in many nations (Henderson et al., 2018). For example, Figure A1-A6 show the nighlights in different continent. Those figures clearly show that brighter areas are located near ports, big rivers or canals. This pattern suggests a strong link between economic activity and international trade, raising a policy-relevant question: Does an exogenous increase in exports raise income predominantly in trade-hub areas, or does it generate more balanced growth across a country's regions?

Answering this question is crucial for two reasons. First, anti-globalization sentiments are increasingly evident in several countries, including the United States, the United Kingdom, and France (Rodrik, 2021). In the 2024 U.S. presidential election, the Democratic candidate secured most votes along the East and West Coasts—regions that have benefited significantly from globalization—while the Republican candidate dominated inland areas farther from coastlines (The Wall Street Journal, 2024; National Association of Counties, 2024). This divergence underscores the uneven distribution of globalization's benefits. While the positive relationship between exports and national income is well established, the localized effects of trade on urban and regional development remain underexplored.

Second, UN projections indicate that by 2050, 68 percent of the global population will reside in urban areas (United Naitons:, 2019). Understanding how globalization affects urban income levels and migration into urban centers is essential for predicting future demographic and economic trends.

Investigating the localized effects of exports on income in trade-hub and non–trade-hub areas is challenging for two reasons. First, measuring economic activity at the subnational level on a global scale is inherently difficult due to limited data availability. Second, exports are endogenous: if an area's productivity increases, both exports and income may rise simultaneously. Moreover, trade openness itself is endogenous—a country may adopt export-promoting policies in response to rising productivity.

We address these challenges by using nightlight intensity and nightlight intensity per capita to proxy for income and income per capita at the subnational level. Nightlight data, available at a fine spatial resolution (1 km times 1 km), allow us to capture variation in economic activity between trade-hub and non–trade-hub areas (Henderson et al., 2012; Martinez, 2022). We construct a subnational panel for 1997–2020 using five-year averages. Trade-hub areas are defined as the union of all locations within a 50 km or 30 km radius of each country's three largest ports and three largest international airports; all other areas are classified as non–trade-hub areas. The dataset covers 101 countries, more than 350 of the world's largest ports, and over 300 inland cities with airports or nearby airports. To avoid mechanically capturing lights from ports and airports themselves, we exclude nighttime luminosity within a 3 km radius of each port and airport.

To address the endogeneity of exports, we employ a time-varying instrument based on the dynamic gravity equation of Feyrer (2019). This instrument exploits exogenous variation in transport costs generated by technological advances that shifted trade from sea to air, interacting year dummies with sea and air transport distances. These changes have altered the effective economic meaning of distance, leading to heterogeneous trade impacts across countries.

Our results show that a 1 percent increase in exports raises nightlight intensity by 0.31 percent in trade-hub areas and by 0.06 percent in non–trade-hub areas, widening the rural–urban income gap by an estimated 15–17 percent between 1997 and 2018. Nightlight intensity per capita rises by 0.19 percent in trade-hub areas and by 0.04 percent in non–trade-hub areas, with modest migration into trade hubs only partially offsetting per capita gains. These findings indicate that export-led growth disproportionately benefits trade-hub areas, and that barriers to migration may limit the spread of these gains to rural regions.

This paper relates to two strands of the literature. First, several papers have examined the effect of trade on regional inequality (Silva and Leichenko, 2004; Meschi and Vivarelli, 2009; Naranpanawa and Arora, 2014; Storeygard, 2016; Hirte et al., 2020). For example, in their study of Sub-Saharan African cities, Storeygard (2016) investigate how a city's distance from a major port influences its income when transportation costs rise. They find that, following an increase in oil prices, cities located near a major port experience a seven percent income gain relative to cities situated 500 kilometers farther inland. Hirte et al. (2020) examine how trade affects within-region inequality.

The second strand of literature focuses on the uneven effects of trade across regions, industries, and population groups (Autor et al., 2016, 2013; Dix-Carneiro and Kovak, 2017; Faber, 2014; Kovak, 2013; Topalova, 2010; Donaldson, 2018). In particular, Autor et al. (2016, 2013) show that rising import competition from China generated uneven adjustment costs across U.S. regions and workers, with trade-exposed areas experiencing larger declines in manufacturing employment, wages, and labor-force participation. Similarly, Dix-Carneiro and Kovak (2017); Kovak (2013); Topalova (2010) document that trade liberalization produced heterogeneous regional effects, partly because limited labor mobility and regional industrial specialization caused trade-exposed regions to experience weaker employment, wage, and poverty outcomes. Faber (2014) investigates that trade integration can raise aggregate welfare, its benefit are unevenly distributed across space, depending on differences in market access and connectivity. Donaldson (2018) presents that expansion of railroad infrastructure in colonial India reduces trade cost, and increase both trade and income at district level.

This paper contributes to the literature in three main ways. First, building on Feyrer (2019), we employ a dynamic gravity framework in which changes in transportation costs generate exogenous variation in trade flows. A central question in the international trade literature concerns the causal impact of trade on economic outcomes. While Feyrer (2019) studies the effects of trade on economic growth, we adapt this dynamic gravity approach to examine the impact of exports on urban–rural inequality. Second, we estimate the model using Poisson Pseudo-Maximum Likelihood (PPML), which naturally accommodates zero trade flows, consistent with the insights of (Silva and Tenreyro, 2006). Third, unlike most existing studies that focus on inequality within regions, we analyze inequality between urban and rural areas. Given the growing debate on trade and inequality, this distinction is economically meaningful and policy relevant.

This paper is organized as follows: Section 2 outlines the estimation strategy. Section 3 presents the results, Section 4 discusses the implications, and Section 5 concludes.

2 Estimation Strategy

2.1 Nighttime Lights and Real GDP

We use nighttime lights data from the United States Air Force Defense Meteorological Satellite Program (DMSP). The dataset combines the original DMSP-OLS nighttime lights series for 1992–2012 with the extended DMSP nighttime lights series released by the National Oceanic and Atmospheric Administration (NOAA) for 2013–2020, following the literature that uses nighttime lights data (Joseph, 2022). The extension series was constructed using pre-dawn nighttime observations from existing DMSP satellites whose orbital crossing times had gradually shifted from evening to dawn hours. The extended data preserve the spatial resolution of the earlier series and allow us to expand the temporal coverage of the analysis. The combined dataset provides annual observations from 1997 to 2020 at a spatial resolution of 1 km times 1 km. As Henderson et al. (2012) demonstrated, nighttime lights data are a strong proxy for gross domestic product (GDP). Storeygard (2016) and Hirte et al. (2020) use nightlight data to examine the effect of trade on regional inequality. From these data, we extract nightlight intensity for both trade-hub and non–trade-hub areas within each country and calculate the total nightlight intensity for each area. For years in which data are available from two satellites, we use the average value. To compute nightlight intensity per capita for trade-hub and non–trade-hub areas, we divide the total nightlight intensity by the local population in each area, as measured using gridded population data.

Because nightlight measurements in regions near the poles are affected by weather and extreme seasonal variation in sunlight, we restrict our sample to a latitude range between -65 degrees and 66.63 degrees, effectively excluding areas north of the Arctic Circle.

Column (1) of Table 1 presents the estimated coefficients from the regression of real GDP on nighttime lights for the set of countries analyzed by Henderson et al. (2018)[1]. Column (2) reports the OLS estimates for our sample of countries. The number of countries in Column (2) is smaller than in Column (1) because a substantial number of countries do not report export values for each trade pair. In both cases, nighttime lights are strongly correlated

[1]Henderson et al. (2018) studied 188 countries, whereas Column (1) of Table 1 uses 187 countries due to the unavailability of data for the former Yugoslav Republic.

with real GDP, controlling for country fixed effects, time fixed effects, and population. The estimated coefficients indicate that a one percent increase in nighttime lights is associated with a 0.2 to 0.3 percent increase in real GDP, even after controlling for country fixed effects, time fixed effects, and population.

Table 1. The Ordinary Least Squares Estiamted Effects of Lights on Real GDP

	(1)	(2)
A. OLS Estimates		
Dependent Variable	Ln Real GDP	
Ln Lights	0.307***	0.260**
	(0.0871)	(0.0998)
Ln Population	0.469***	0.567***
	(0.0940)	(0.126)
R-squared	0.998	0.999
B. Specifications		
Country FE	Yes	Yes
Time FE	Yes	Yes
Countries	187	187
N	1106	1106

Notes: The nightlights and real GDP data with five-years is used for the regression estimation from 1992 to 2020. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

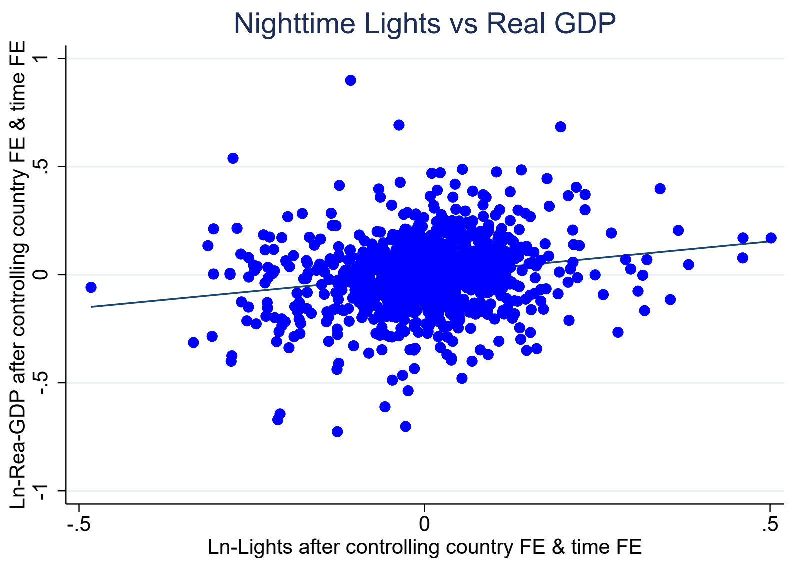
Figure 1 displays a scatter plot of the residual relationship between nighttime lights and real GDP, after controlling for country-fixed effects and time-fixed effects and population.[2]

Both Figure 1 and Table 1 indicate that nighttime luminosity is strongly correlated with real GDP or real GDP per capita, even after accounting for country and time fixed effects.[3]

[2] In this figure, the natural logarithm of real GDP is first regressed on control variables, country fixed effects, and time fixed effects to obtain the residuals. The same procedure is applied to the natural logarithm of nighttime lights. The scatter plot is then constructed using the residuals from these regressions.

[3] In addition, the robustness check section will demonstrate that nighttime lights are strongly correlated with household welfare, using data from the Demographic and Health Surveys. These findings suggest that nighttime lights serve as a reliable proxy for local economic activity, allowing researchers to examine the effects of exports on economic outcomes.

Figure 1: The Association between Nighttime Lights and Real GDP



Notes: The average nighttime lights data from DMSP are used for the period 1997–2020. The data are presented in five-year intervals to plot this graph.

2.2 Main equation

We hypothesize that exports increase income and income per capita in trade-related areas (trade-hub areas) more than in non–trade-related areas (non–trade-hub areas). To test this hypothesis, we estimate the following equation:

$$Y^a_{jt} = \alpha_0 + \alpha_1 \ln(\text{Exports}_{jt}) + \alpha_2 \mathbf{Z}_{jt} + \eta_j + \eta_t + \epsilon_{jt} \tag{1}$$

j indexes the source country and t indexes the time period, measured in five-year intervals. For example, $t = 1$ corresponds to 1997–2000, $t = 2$ corresponds to 2001–2005 and $t = 3$ corresponds to 2006–2010. The superscript a denotes whether the outcome pertains to a trade-hub area or a non–trade-hub area. A trade-hub area is defined as the union of areas within a 50 km or 30 km radius of each of the three largest ports and three largest international airports in a country. A more detailed definition is provided in Section 2.4.1.

The dependent variable Y^a_{jt} is the natural logarithm of (i) the five-year average of nightlight intensity, (ii) nightlight intensity per capita, (iii) the local population ratio, or (iv) the local population, for either trade-hub or non–trade-hub areas. The local population ratio is defined as the share of the national population residing in a given area type.

The key explanatory variable $\ln(\text{Exports}_{jt})$ is the natural logarithm of five-year-averaged exports. The control vector $\mathbf{Z}_{jt}$ includes national population and local weather conditions (temperature and precipitation). National population is included because, in some specifications, the dependent variable is nightlight intensity per capita, calculated as total nightlight intensity divided by the local (trade-hub or non–trade-hub) population. Weather variables are included because weather conditions may affect both exports and nightlight intensity.

Country fixed effects (η_j) control for unobserved, time-invariant country-specific factors influencing economic growth, such as institutional quality, culture, and baseline development levels. Time fixed effects (η_t) control for common shocks and global trends. The error term ϵ_{jt} allows for serial correlation over time.

2.3 Gravity Equation

To estimate equation (1), a simple OLS regression does not capture the causal effect because exports are an endogenous variable. For example, a country with higher productivity may enjoy higher income and consequently export more. Additionally, a country's openness may be a policy target driven by its competitiveness. Thus, it is necessary to use an instrumental-variable approach to address the endogeneity of export volume.

In one of the most frequently cited papers on the effect of trade on income, Frankel and Romer (1999) employs geographic factors to predict trade in cross-sectional data and uses the predicted trade as an instrument to examine the effect of trade on income. However, the instrument introduced by Frankel and Romer (1999) may not satisfy the exclusion restriction because countries possess different geographic characteristics that can influence income through channels other than trade. For instance, countries located near the equator are generally farther from major international markets than others, and those with unfavorable disease environments or unproductive colonial institutions may engage in less trade.

Feyrer (2019) introduces a time-varying instrumental variable derived from a gravity equation. The instrument is constructed from predicted export values estimated by interacting sea distance and air distance with year dummies in a regression on actual exports. These predicted values capture temporal changes in trade patterns, particularly the shift from sea freight to air cargo. By employing this time-varying instrument in panel regressions, it becomes possible to control for country-specific factors influencing economic growth, thereby mitigating omitted variable bias more effectively than in cross-sectional analyses.

We extend the instrumental variable proposed by Feyrer (2019) and construct our own predicted export values. First, we identify the largest airport in each country and define air distance as the direct point-to-point distance between the largest airports of two countries. Similarly, bilateral sea distance is manually computed using raw geographical data by identifying the largest port in each country and then calculating the sea distance between the two ports in a country pair.[4]

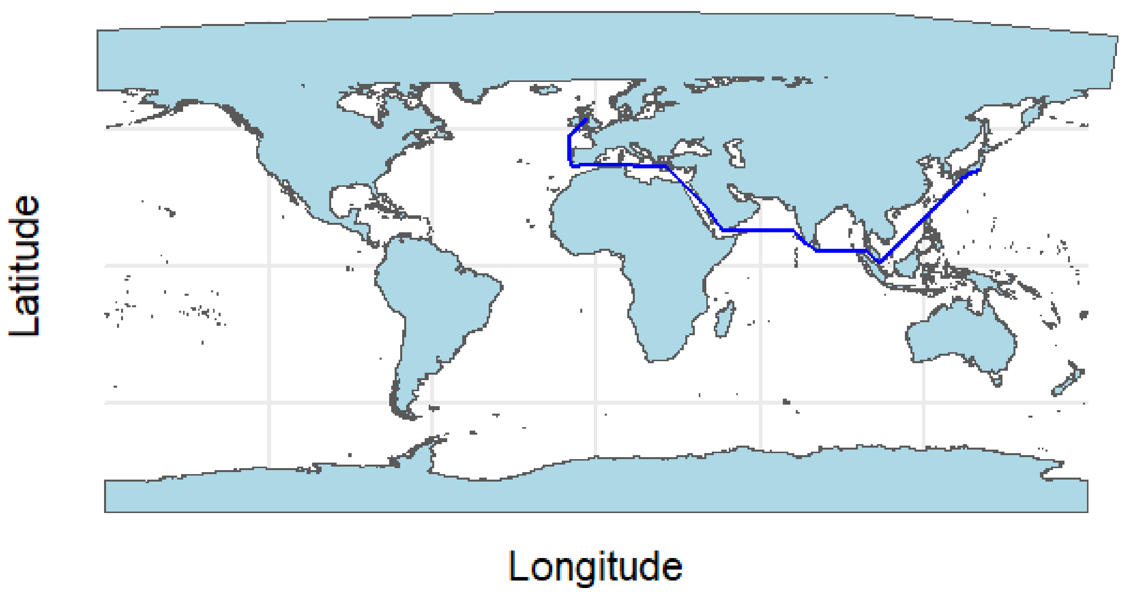
In addition, following the spirit of Feyrer (2019), we include surface distance interacted with year dummies and the number of shared borders interacted with year dummies in the gravity equation. The surface distance is the point-to-point distance between pairs of countries located on the same continent and it is calculated between the capital cities for each source-destination pair. Surface distance is longer for countries with greater distances on landmasses and incurs higher land transportation costs. We assume that Surface distance is infinity for pair of countries that are not located in the same continent. The number of shared borders indicates how many countries a given country shares land borders with.

The sea distance from the United States and Canada to other countries can differ depending on whether it is the west coast or east coast. In the United States and Canada, the west and east coasts serve as independent economic entities, and the sea distances to these coasts are significantly different, which could affect the results in obtaining the instrument. For the United States and Canada, the average sea distance to two ports—one from the east coast and one from the west coast—is calculated. In the United States, the Port of New York/New Jersey on the east coast and the Port of Los Angeles on the west coast were selected, while in

[4]To calculate sea distance, we removed land from a global raster map by assigning a value of 1 to sea and 0 to land, then computed the shortest sea path between countries. Both air and sea distances are measured in thousands of kilometers.

Canada, the Port of Montreal on the east coast and the Port of Vancouver on the west coast were selected. This may address concerns about differences in sea distances between the east and west coasts of the United States and Canada. In another gravity equation, we used the sea distance from either the east coast or the west coast alone. The results using the sea distance from ports on different coasts in the United States and Canada are consistent with the main findings. Even when we excluded the United States and Canada from the sample, the regression estimations remained consistent with the main results and the results are available on Supplemental Information C. Figure 2 describes the calculation of the shortest path from Liverpool port in United Kingdom to Tokyo port in Japan. [5]

Figure 2: Shortest Sea Path from Liverpool Port in United Kingdom to Tokyo Port in Japan



Source: Calculated by Authors

We utilize annual export pair data for the gravity equation to obtain the predicted exports. The export pair data contains several zeros across different pairs as well as across years. Zero export values are possible for trade partners, especially in the case of developing

[5]We expanded the size of the Suez Canal properly so that the programming algorithm can find the path properly through the Suez Canal if it is the shortest shipping route at sea.

countries. In the literature, most previous papers including Frankel and Romer (1999) used a log-linearized model. However, such a model, in the presence of heteroskedasticity, provides biased and inconsistent estimates of parameters (Silva and Tenreyro, 2006).

Thus, following the suggestion by Silva and Tenreyro (2006), we use the Poisson Pseudo-Maximum Likelihood (PPML) estimation, incorporating exports with zero values. Such a non-linear model appropriately handles the zero values of exports and provides a consistent, unbiased estimates.

To implement PPML, we assume that the following relationship holds:

$$\begin{aligned}\mathbb{E}(\text{Exports}_{jiy}|X_{jiy}) = \exp\Big(&\alpha_0 + \theta_{ji} + \theta_y + \gamma_{sea,y} \times \ln(\text{Sea Distance}_{ji}) \\ &+\gamma_{air,y} \times \ln(\text{Air Distance}_{ji}) + \gamma_{surface,y} \times \ln(1 + 1/\text{Surface Distance}_{ji}) \\ &+\gamma_{sharedborder,y} \times \text{Shared Land Border}_{ji}\Big) \end{aligned} \tag{2}$$

where j, i and y denote the source country, target country, and year, respectively. Note that in the gravity equation, we use annual data in order to increase the predictive power to predict export through our instrumental variables. The above equation shows that the export is a function of air distance, sea distance, surface distance and sharing border dummy. The presence of $\gamma_{air,y}$, $\gamma_{sea,y}$, $\gamma_{surface,y}$ and $\gamma_{sharedborder,y}$ means that the effect of sea distance and other related variables have different effects at different year. In estimation, this can be handled by interacting distance variables with year dummies. The outcome variable is the annual export of source country j to target country i at year y. X_{ijy} is a set of explanatory variables such as sea distance, air distance, shared Land border dummies, their interaction with year dummies. θ_{ij} is the pair fixed effect, and θ_y is the year fixed effect.

The set of parameters in the above model is obtained by solving the following condition (Motta, 2019):

$$\sum_{j,i,y} \left(\text{Exports}_{jiy} - \mathbb{E}(\text{Exports}_{jiy}|X_{jiy})\right) X_{jiy} = 0 \tag{3}$$

where X_{jiy} are the explanatory variables in the gravity equation.

Once the estimated coefficients $\hat{\gamma}$ are obtained, the predicted export values, $\widehat{\text{Exports}}_{ijt}$, can be obtained by plugging the estimated coefficients back into the exponential functional form:

$$\widehat{\text{Exports}}_{jiy} = \exp\Big(\hat{\alpha}_0 + \hat{\theta}_{ji} + \hat{\theta}_y + \hat{\gamma}_{sea,y} \times \ln(\text{Sea Distance}_{ji}) + \hat{\gamma}_{air,y} \times \ln(\text{Air Distance}_{ji})$$
$$+\hat{\gamma}_{surface,y} \times \ln(1 + 1/\text{Surface Distance}_{ji}) + \hat{\gamma}_{sharedborder,y} \times \text{Shared Land Border}_{ij}\Big) \quad (4)$$

These predicted values represent the expected exports between country pairs given the model's explanatory variables. We then summed the predicted exports to obtain the total predicted exports for each source country and each period:

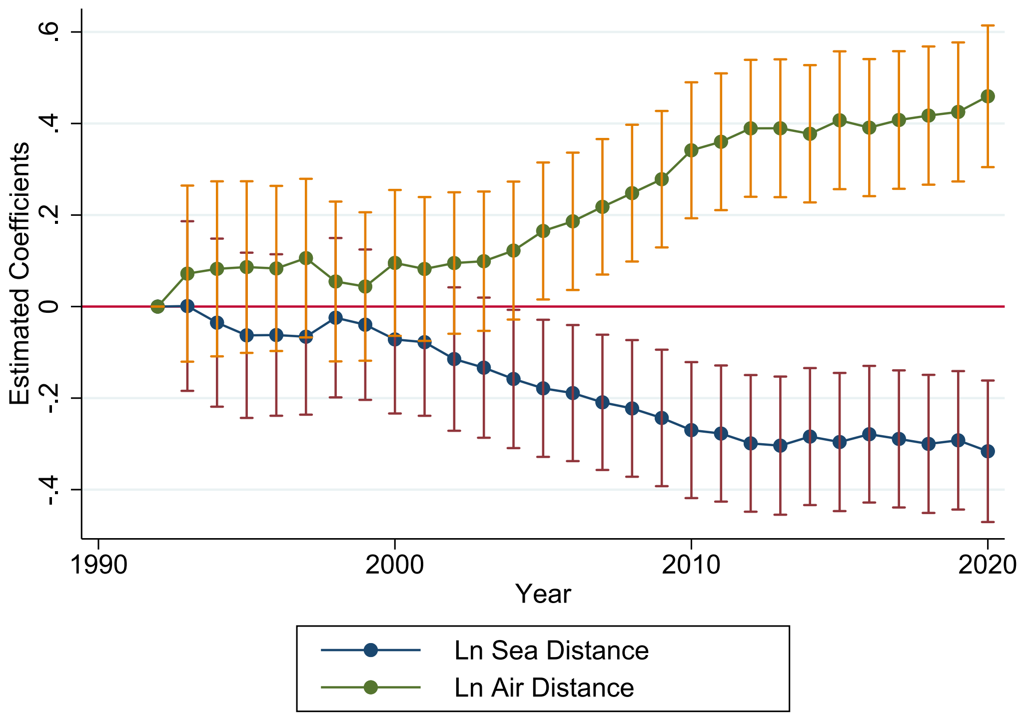
$$\text{Predicted Exports}_{jt} = \sum_{y \in t} \sum_i \widehat{\text{Exports}}_{ijy} \quad (5)$$

$$\ln \mathrm{E}xports_{jt} = \delta_0 + \delta_1 \ln \text{Predicted Exports}_{jt} + \delta_2 \mathbf{z}_{jt} + \lambda_j + \lambda_t + \varepsilon_{jt} \quad (6)$$
$$Y^a_{jt} = \alpha_0 + \alpha_1 \ln \text{Exports}_{jt} + \alpha_2 \mathbf{z}_{jt} + \eta_j + \eta_t + \xi_{jt} \quad (7)$$
$$j = 1, 2, \ldots, 101, \; t = 1, 2, \ldots, 5.$$

where j denotes the country and t denotes the time period, measured in five-year intervals. $y \in t$ denotes all years included in period t. Equation (6) represents the first-stage estimation. The endogenous explanatory variable is Ln Exports$_{jt}$, which represents the natural logarithm of the total exports of country j at time t. The instrumental variable for this endogenous variable is Ln Predicted Exports$_{jt}$, the natural logarithm of predicted exports, which are estimated based on changes in geographic advantages.

Figure 3 shows the estimated coefficients for natural logarithm of air distance and sea distance by estimating equation (2) for our data set over the span of 29 years, from 1992 to 2020. The estimated coefficients of natural logarithm of air distance over time indicate the marginal effect of longer air distance become more positive to export as time passes and the marginal effect of sea distance become more negative as time passes. It shows that shipping through air cargo become more beneficial and shipping through sea became more

Figure 3: Changing Importance of Different Transportation Mode: The Estimated Coefficients of Sea Distance and Air Distance over Time



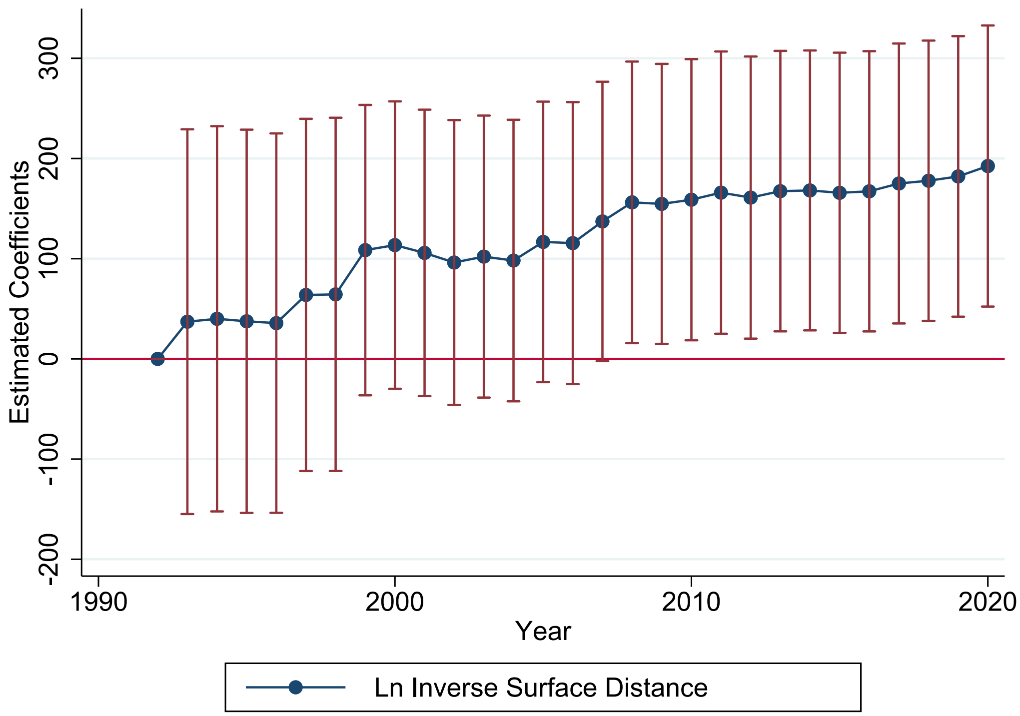
Notes: This figure is based on the estimation results from a Poisson Pseudo-Maximum Likelihood (PPML) model using equation (2). The dependent variable is exports, and the explanatory variables include distance measures, their interactions with year dummies, and fixed effects. To create the graph, the estimated coefficients and standard errors for the natural logarithm of sea distance by year and air distance by year are presented with 95% confidence intervals.

disadvantageous.

Figure 4 describes the change in export levels with respect to one plus the reciprocity of surface distance. Note that if two countries are on different continents, we assume that surface distance is infinity and $\ln(1 + 1/\text{surface distance}_{ij}) = 0$. Essentially, Figures 4 and 3 show the same pattern. Holding air distance and sea distance constant, a shorter surface distance is correlated with an increase in relative exports over time.

Figure 5 shows the change in the marginal effect of having borders with many countries.

Figure 4: The Change in Export Level with Respect to Surface Distance in the Same Continent Over Time From a Gravity Regression With Pair Fixed Effect



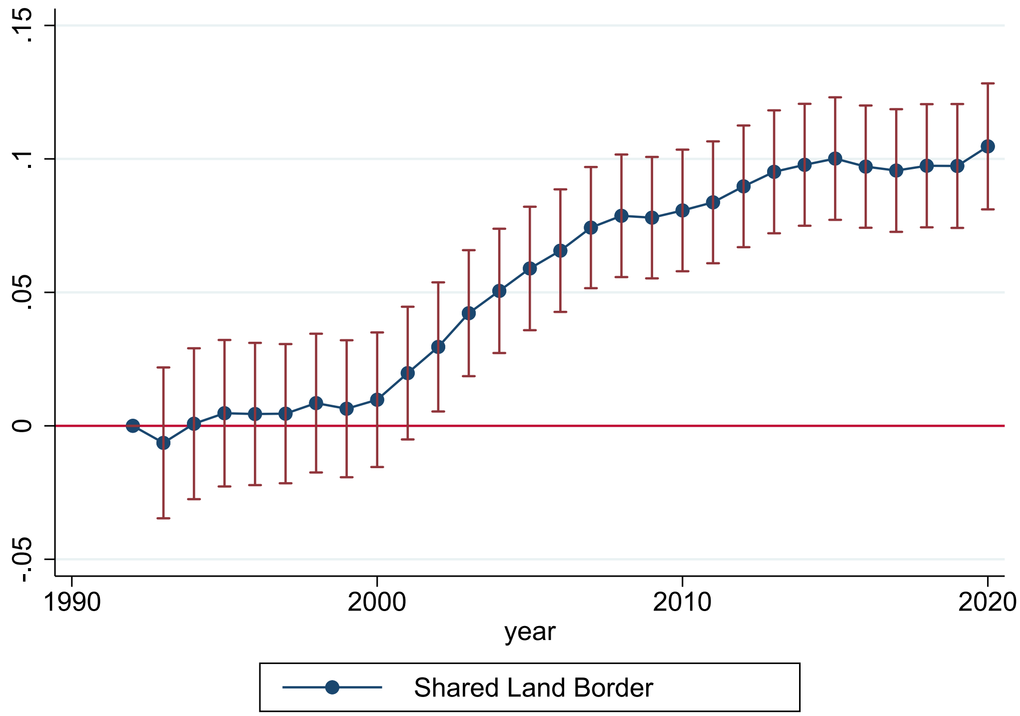
Notes: This figure is based on the estimation results from a Poisson Pseudo-Maximum Likelihood (PPML) model using equation (2). The dependent variable is exports, and the explanatory variables distance measures, their interaction with year dummies, and fixed effects. To create the graph, the estimated coefficients and standard errors for the $\ln(1 + 1/\text{Surface Distance}_{ij})$ by year is presented with 95% confidence intervals.

The explanatory variable is the number of shared borders. Figure 5 indicates that it is more advantageous to share borders with many countries. Due to the very short distances with neighboring countries, conducting exports over land has been much more effective in such cases, which might be attributed to the construction of highway networks across countries.

2.4 Dataset Construction

We select countries which have at least on port and we exclude landlocked countries. For the choice of countries, we follow the guideline practiced by Henderson et al. (2012). We do not include Serbia and Montenegro in the sample due to border changes, and we exclude

Figure 5: The Change in Export Level with Respect to Shared Land Border Over Time From a Gravity Regression With Pair Fixed Effect



Notes: This figure is based on the estimation results from a Poisson Pseudo-Maximum Likelihood (PPML) model using equation (2). The dependent variable is exports, and the explanatory variables are distance measures, their interaction with year dummies, and fixed effects. To create the graph, the estimated coefficients and standard errors for the shared land border by year are presented with 95% confidence intervals.

small island countries. Bahrain, Hong Kong, and Qatar are omitted because of top-coded lights and the lack of a non-trade hub area. Additionally, we exclude Equatorial Guinea, as most of its lights come from gas flares. Finally, following the above procedure, we select 101 countries.

To measure exports, bilateral trade data from 1997 to 2020 were obtained from the Direction of Trade (DoT) database provided by the International Monetary Fund (IMF). Since exports are unlikely to have an immediate effect on nightlight intensity, we use five-year averages of export values in our analysis. Moreover, annual nightlight data can be noisy, and aggregating the data at five-year intervals helps reduce this noise and mitigate distortions.

The fourth version of the Gridded Population of the World (GPWv4) dataset was obtained from the Socioeconomic Data and Applications Center (SEDAC) Center for International Earth Science Information Network–CIESIN–Columbia University (2018). This spatially disaggregated dataset provides population counts at five-year intervals—for the years 2000, 2005, 2010, 2015, and 2020—with a resolution of 30 arc-seconds (approximately one kilometer at the equator). The local population count (number of persons per pixel) for both trade hub and non–trade hub areas is extracted using the same spatial procedure applied to the night lights data.

Temperature and precipitation data are obtained from the Climate Research Unit Time Series (CRU-TS) version 4.06 (Harris et al., 2020). This dataset provides monthly observations from 1901 to 2020 at a spatial resolution of $0.5° \times 0.5°$ (latitude/longitude) grids. For our analysis, the data were aggregated into five-year averages at the local level for both trade hub and non–trade hub areas, as well as at the national level.

2.4.1 Trade-Hub and Non–Trade-Hub Areas

Trade-hub areas are defined as the union of areas within a 50 km (main specification) or 30 km (robustness check) radius of each of the three largest ports (port urban areas) and three largest international airports (airport urban areas) in a country. The robustness check confirms that the main results are robust to reducing the radius from 50 km to 30 km.

To identify port locations, we obtained geographic coordinates of coastal and inland ports from Google Earth. A buffer area with a 50 km or 30 km radius was then constructed around each port city and inland city. The trade-hub area for each country is defined as the union of its three port urban areas and three airport urban areas. If a country has fewer than three ports, those available ports are included. Ports used primarily for non-trade purposes, such as fisheries or tourism, are excluded.

In total, our sample includes over 350 of the world's largest ports and more than 300 inland cities with major airports. The non–trade-hub area for each country is defined as the remaining territory after removing the trade-hub areas from the national boundary.

3 Results

3.1 Main Results

Table 2 presents the descriptive statistics of the variables. Panels A, B, and C report the statistics for the trade-hub area, the non–trade-hub area, and the entire country at five-year intervals, respectively. The table shows that the median nightlight per area in trade-hub areas is 12.27, compared with 4.71 in non–trade-hub areas. Median nightlight per capita per area is 0.018 in trade-hub areas and 0.006 in non–trade-hub areas. The median population share in trade-hub areas is 0.45, compared with 0.55 in non–trade-hub areas, while trade-hub areas account for only 19 percent of national land area. These statistics indicate that economic activity per unit of area is higher in urban areas, even though the population is relatively evenly distributed between urban and rural areas.

Table 2. Descriptive Statistics

Variable	Mean	Median	Std. Dev.	Min	Max
A. Trade Hub Area					
Lights per Area	16.244	12.27	13.884	2.184	64.813
Ln Lights per Area	2.454	2.51	0.825	0.781	4.172
Lights per Capita per Area ×10K	0.055	0.018	0.124	0.001	1.366
Ln Lights per Capita per Area	-13.1	-13.21	1.276	-16.035	-8.898
Population (in million)	11.984	6.42	16.199	0.193	125.099
Population Ratio	0.458	0.42	0.226	0.036	0.992
Temprature (°C)	19.581	22.28	7.135	2.667	28.471
Precipitation (mm)	103.682	95.18	62.733	5.19	292.441
Share of Trade Hub Area	0.187	0.10	0.216	0.001	0.952
B. Non-Trade Hub Area					
Lights per Area	7.144	4.71	5.973	1.992	40.178
Ln Lights per Area	1.741	1.55	0.619	0.689	3.693
Lights per Capita per Area × 10K	0.099	0.006	0.00003	0.00004	2.247
Ln Lights per Capita per Area	-13.974	-14.32	2.228	-19.386	-8.401
Population (in million)	48.667	7.68	169.281	0.02	1366.414
Population Ratio	0.542	0.58	0.226	0.008	0.964
Temprature (°C)	18.964	22.59	8.069	-3.339	29.023
Precipitation (mm)	105.159	89.07	68.572	2.152	331.763
C. Country Level Variables					
Ln Export	-4.444	-4.52	2.367	-12.614	0.838
Ln Predicted Exports	-4.369	-4.56	2.274	-10.857	0.756
Difference of ln Lights	-1.412	-1.51	1.474	-4.935	3.989
Difference of ln Lights per Capita	-1.25	-1.16	0.776	-3.697	0.267
Difference of ln Population	-0.162	-0.33	1.199	-3.278	4.848
Lights per Area	8.352	5.41	7.553	2.06	52.545
Ln Lights per Area	1.865	1.69	0.664	0.723	3.962
Population (in million)	60.652	14.70	181.416	0.253	1443.72
Temprature (°C)	18.999	22.58	7.998	-2.845	28.875
Precipitation (mm)	103.126	90.29	65.995	2.202	318.971
Number of Observations			505		
Number of Countries			101		

Note: Data is in five-years interval from 1997 to 2020.

Table 3 presents the Ordinary Least Squares (OLS) results. The dependent variable is the natural logarithm of nightlight intensity in each area (trade hub or non–trade hub), and the main explanatory variable is the natural logarithm of exports. Columns (2) and (5) control for national population to capture the effect on nightlight intensity per capita. Columns (3) and (6) add controls for local weather conditions, which can directly influence nightlight measurements through effects on electricity generation and cloud cover.

Table 3. Ordinary Least Squares Estimated Effects of Exports on Lights in Trade Hub Area and Non-Trade Hub Area

	(1)	(2)	(3)	(4)	(5)	(6)
Area	Trade Hub Areas			Non-Trade Hub Areas		
A. OLS Estimates						
Dependent Variable	Ln Lights			Ln Lights		
ln Exports	0.116***	0.116***	0.117***	0.0446*	0.0457**	0.0441*
	(0.0385)	(0.0383)	(0.0391)	(0.0235)	(0.0228)	(0.0229)
R-Squared	0.988	0.988	0.988	0.998	0.998	0.998
B. Specifications						
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
National Population		Yes	Yes		Yes	Yes
Local Weather			Yes			Yes
Countries	101	101	101	101	101	101
N	505	505	505	505	505	505

Notes: The control variables include country-level population and local weather for trade hub and non-trade hub areas. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

Table 4 reports the first-stage results of the Two-Stage Least Squares (2SLS) estimation. Predicted exports, derived from our dynamic gravity equation, are strongly correlated with actual exports. The Kleibergen–Paap Wald F-statistic is 31.47, exceeding the conventional threshold of 10, indicating a strong instrument. Following Windmeijer (2025), the effective F-statistic for a single endogenous variable is 31.75—again above the standard cutoff—further confirming instrument strength.

Table 5 presents the second-stage and reduced-form regression results from the 2SLS estimation. The estimated coefficients in Columns (3) and (6) indicate that a one percent increase in exports raises nightlight intensity by 0.31 percent in trade hub areas and 0.06 percent in non–trade hub areas, respectively.

One potential concern is that the analysis compares highly urbanized areas (trade hubs)

Table 4. First Stage Estimated Effects of Predicted Exports on Actual Exports

	(1)	(2)	(3)
A. OLS Estimates			
Dependent Variable		Ln Exports	
ln Predicted Exports	1.006***	1.004***	0.986***
	(0.155)	(0.154)	(0.157)
Kleibergen–Paap rk Wald F Statistic	33.47	34.05	31.47
Effetive F Statistic	33.74	34.32	31.75
R-Squared	0.983	0.983	0.983
B. Specifications			
Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
National Population		Yes	Yes
Local Weather			Yes
Countries	101	101	101
N	505	505	505

Notes: The control variables include country-level population and weather. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

with very remote regions, potentially exaggerating disparities. To address this, we estimate a 2SLS model using national-level nightlight intensity as the dependent variable. Table 6 presents the results. The estimated coefficients on $\ln(\text{Export})$ are very similar to those for non–trade hub areas in Table 5, suggesting that the national-level effect largely reflects the impact on non–trade hub areas. This is reasonable, as trade-hub areas are defined as the union of all locations within 50 km of an international port or airport, accounting for only 19 percent of the national land area.

Table A2 presents the second-stage and reduced-form results of the 2SLS estimation, where the dependent variable is the difference between the natural logarithm of nightlight intensity in trade hub and non–trade hub areas.[6] The results indicate that a one percent increase in exports raises this difference by 0.25 percentage points. Consistent with Table 5, this provides further evidence that export growth widens the income gap between urban and

[6] See Table A2 in the Supplementary Information section.

Table 5. The Second Stage Estimated Effects of Exports on Lights in Trade Hub Area and Non-Trade Hub Area

	(1)	(2)	(3)	(4)	(5)	(6)
Area	Trade Hub Area			Non-Trade Hub Area		
A. Second-Stage Estimates						
Dependent Variable	ln Lights			ln Lights		
ln Exports	0.300***	0.301***	0.305***	0.0638	0.0683	0.0613
	(0.0660)	(0.0648)	(0.0658)	(0.0785)	(0.0782)	(0.0818)
B. Reduced Form Estimates						
ln Predicted Exports	0.302***	0.303***	0.301***	0.0641	0.0686	0.0601
	(0.0804)	(0.0796)	(0.0797)	(0.0844)	(0.0845)	(0.0857)
R-Squared	0.989	0.989	0.989	0.998	0.998	0.998
C. Specifications						
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
National Population		Yes	Yes		Yes	Yes
Local Weather			Yes			Yes
Countries	101	101	101	101	101	101
N	505	505	505	505	505	505

Notes: The control variables include country-level population and local weather for trade hub and non-trade hub areas. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

rural areas.

The findings in Tables 5 and A2 naturally raise the question of whether the widening gap in nightlight intensity induces migration from rural to urban areas. Table 7 examines the effect of exports on the local population share using a 2SLS estimation, while Table A3 analyzes the effect on the difference in the logarithm of population between the two area types.[7] The results in Table 7 indicate that a one percent increase in exports raises the population share in trade-hub areas by 0.012 percentage points and reduces it in non–trade hub areas by the same amount. Given population shares of 0.45 and 0.55, respectively, this corresponds to a 0.027 percent increase in trade-hub population and a 0.022 percent

[7] See Table A3 in the Supplementary Information section.

Table 6. The Second Stage Estimated Effects of Exports on Lights: Regression Estimations at the Country Level

	(1)	(2)	(3)
A. Second-Stage Estimates			
Dependent Variable		ln Lights	
Ln Exports	0.0647	0.0697	0.0645
	(0.0782)	(0.0780)	(0.0802)
B. Reduced Form Estimates			
Ln Predicted Exports	0.0651	0.0700	0.0634
	(0.0842)	(0.0845)	(0.0846)
R-Squared	0.997	0.997	0.997
C. Specifications			
Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
National Population		Yes	Yes
National Weather			Yes
Countries	101	101	101
N	505	505	505

Notes: The control variables include country-level population and weather. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

decrease in non–trade hub population. Using the population-log difference as the dependent variable, Table A3 shows that a one percent increase in exports widens the population gap by 0.09 percent. These results suggest that exports promote rural–urban migration, although the magnitude is small relative to the impact of exports on income, as proxied by nightlight intensity.

Table 8 presents the second-stage and reduced-form estimates of the effect of exports on nightlight intensity per capita. The coefficients in Columns (3) and (6) indicate that a one percent increase in exports raises lights per capita by 0.19 percent in trade-hub areas and by 0.04 percent in non–trade hub areas. Table 9 reports the corresponding national-level results, showing that the estimated coefficient on $\ln(\text{Exports})$ is very similar to that for non–trade hub areas in Table 8. This suggests that the national-level effect primarily reflects the impact on non–trade hub areas, while the effect in trade-hub areas is substantially larger. These results

Table 7. The Second Stage Estimated Effects of Exports on Population Ratio in Trade Hub Area and Non-Trade Hub Area

	(1)	(2)	(3)	(4)	(5)	(6)
Area	Trade Hub Area			Non-Trade Hub Area		
A. Second-Stage Estimates						
Dependent Variable	Population Ratio			Population Ratio		
ln Exports	0.00992	0.00974	0.0125	-0.00992	-0.00974	-0.0125
	(0.0132)	(0.0129)	(0.0150)	(0.0132)	(0.0129)	(0.0150)
B. Reduced Form Estimates						
ln Predicted Exports	0.00997	0.00978	0.0123	-0.00997	-0.00978	-0.0123
	(0.0133)	(0.0129)	(0.0148)	(0.0133)	(0.0129)	(0.0148)
R-Squared	0.993	0.993	0.994	0.993	0.993	0.994
C. Specifications						
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
National Population		Yes	Yes		Yes	Yes
Local Weather			Yes			Yes
Countries	101	101	101	101	101	101
N	505	505	505	505	505	505

Notes: The control variables include country-level population and local weather for trade hub and non-trade hub areas. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

imply that individuals in trade-hub areas benefit more from export growth, resulting in higher per capita income. Supporting this interpretation, Table A4 shows that a one percent increase in exports raises the difference in the logarithm of lights per capita between trade hub and non–trade hub areas by 0.15 percent.[8]

[8]See Table A4 in the Supplementary Information section.

Table 8. The Second Stage Estimated Effects of Exports on Lights per Capita in Trade Hub Area and Non-Trade Hub Area

	(1)	(2)	(3)	(4)	(5)	(6)
Area	Trade Hub Area			Non-Trade Hub Area		
A. Second-Stage Estimates						
Dependent Variable	ln Lights per Capita			ln Lights per Capita		
ln Exports	0.165	0.196***	0.192***	0.0183	0.0494	0.0395
	(0.105)	(0.0658)	(0.0678)	(0.119)	(0.0840)	(0.0908)
B. Reduced Form Estimates						
ln Predicted Exports	0.166	0.197**	0.190**	0.0184	0.0686	0.0388
	(0.110)	(0.0764)	(0.0766)	(0.122)	(0.0845)	(0.0930)
R-Squared	0.977	0.986	0.986	0.988	0.998	0.993
C. Specifications						
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
National Population		Yes	Yes		Yes	Yes
Local Weather			Yes			Yes
Countries	101	101	101	101	101	101
N	505	505	505	505	505	505

Notes: The control variables include country-level population and local weather for trade hub and non-trade hub areas. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

Table 9. The Second Stage Estimated Effects of Exports on Lights per Capita: Regression Estimations at the Country Level

	(1)	(2)	(3)
A. Second-Stage Estimates			
Dependent Variable		ln Lights per Capita	
Ln Exports	0.0354	0.0697	0.0645
	(0.119)	(0.0780)	(0.0802)
B. Reduced Form Estimates			
Ln Predicted Exports	0.0356	0.0700	0.0634
	(0.122)	(0.0845)	(0.0846)
R-Squared	0.985	0.993	0.993
C. Specifications			
Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
National Population		Yes	Yes
National Weather			Yes
Countries	101	101	101
N	505	505	505

Notes: The control variables include country-level population and weather. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

3.2 Does Nighttime Light Capture Household Living Standards?

In the previous subsection, we used nightlight intensity as proxy for income. However, several studies have questioned the validity of nightlight intensity as a measure of income or household living standards at the subnational level in low-income countries (Keola et al., 2015; Gibson et al., 2021). This concern is particularly relevant in the context of Sub-Saharan Africa, where many regions exhibit little or no light emissions. To evaluate whether nightlight intensity per capita accurately reflects household living standards, we examined its relationship with data from the Demographic and Health Surveys (DHS). Household living standards were proxied using indicators of asset ownership and housing conditions, including access to electricity, use of clean cooking fuel, type of flooring material, and ownership of a radio, television, bicycle, and motorcycle.

Since the Demographic and Health Survey (DHS) is not a panel dataset, we construct a quasi-panel using the GPS coordinates of each cluster (village). Specifically, we designate the earliest available survey as the base wave and match each cluster in subsequent waves to a cluster in the base wave if it is located within 10 kilometers. If multiple clusters in a subsequent wave fall within this distance, we select the one closest to the base cluster. Clusters in the base wave with no corresponding match in later waves are excluded from the analysis. Our sample is limited to Sub-Saharan African countries with at least two DHS survey waves that include GPS data.[9]

After constructing the quasi-panel dataset of clusters, we create a 10-kilometer buffer around each cluster in the base wave and calculate the average nightlight intensity for each five-year period. The variable period denotes these five-year intervals.

To quantify the effect of nightlight intensity on household living standards, we estimate the following equation:

$$\mathrm{Y}_{cth} = \beta_0 + \beta_1 \ln(\mathrm{Lights}_{ct}) + \beta_2 \mathrm{z}_{cth} + \lambda_c + \lambda_t + \eta_{cth} \tag{8}$$

[9] Table A9 in the Supplementary Information lists the Sub-Saharan African countries and the DHS waves included in the analysis.

In this specification, c and t denote the cluster and time period indices, respectively, with time measured in five-year intervals. The subscript h denotes the household. We compile DHS data from 17 countries and construct a quasi-panel dataset based on GPS coordinates, assigning each cluster a unique index c. The outcome variable, Y_{cth}, is a binary indicator representing asset ownership or housing conditions. The key explanatory variable, $\ln(\text{Lights}_{ct})$, is the natural logarithm of the five-year average of nightlight intensity in each cluster. The vector $\mathbf{z}_{cth}$ includes household-level control variables such as the household head's years of schooling, age, and gender; the mother's years of schooling and age; and household size.

We control for cluster fixed effects, λ_c, which account for unobserved, time-invariant characteristics specific to each cluster, and time fixed effects, λ_t, which capture period-specific shocks common to all clusters. The error term is denoted by η_{cth}. Standard errors are clustered at the cluster level to allow for correlation of error terms within clusters across households and time periods.

Table 10 presents the estimated coefficients, standard errors, and R-squared values. We run the regressions separately for each outcome variable.[10] The results consistently show that nightlight intensity have a positive and statistically significant effect on household asset ownership and housing conditions.

[10]Descriptive statistics are provided in Table A8 in Supplementary Information A.

Table 10. Estimated Effects of Nighttime Lights on Household Living Standards

	(1) Estimated Coefficient	(2) S. E	(3) R-Squared
List of Dependent Variables			
Electricity Access	0.0643***	(0.0133)	0.600
Radio Ownership	0.158***	(0.0102)	0.193
Television Ownership	0.0914***	(0.0101)	0.455
Bicycle Ownership	0.0300***	(0.00861)	0.323
Motorcycle Ownership	0.0388***	(0.00486)	0.342
Improved Drinking Water Source	0.0978***	(0.0209)	0.364
Improved Cooking Fuel	0.0570***	(0.00786)	0.536
Improved Sanitation Facility	0.0699***	(0.00971)	0.546
Good Floor Materials Ownership	0.0117	(0.0131)	0.484
Cluster FE		Yes	
Time FE		Yes	
Explanatory Variables		Ln Lights and Controls	
Countries		18	
Clusters		5,221	
N		272,455	

Notes: We ran separate regressions for each outcome variable. Controls include household head characteristics and household size. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each cluster. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

3.3 Robustness Checks

In the previous subsection, we showed that an increase in exports leads to higher nightlight intensity in trade-hub areas. However, one might be concerned that defining trade-hub areas as the union of areas within a 50 km radius of each of the three major ports and three international airports in a country could primarily capture lights from the ports or airports themselves, rather than from surrounding economic activity. To address this concern, we redefine trade-hub areas as the union of areas within a 50 km radius of each of the three major ports and international airports, but exclude lights within a 3 km radius of each port and airport from the analysis. This adjustment ensures that lights emitted directly by ports and airports are excluded, allowing us to better capture surrounding economic activity within trade-hub areas.

Tables 11, 12, and 13 show that even after excluding lights within a 3 km radius of each port and airport, the results remain consistent with the main findings. The estimated effects of exports on the differences in nightlight intensity, population, and lights per capita are provided in the Supplementary Information section (see Tables A5, A6, and A7 in Supplementary Information A).

As a second robustness check, we redefine trade-hub areas as the union of areas within a 30 km radius of each of the three major ports and three international airports. The estimated effects of exports on nightlight intensity, population, and lights per capita are reported in the Supplementary Information section (see Tables B1–B7 in Supplementary Information B). The results indicate that a one percent increase in exports raises nightlight intensity by 0.31 percent in trade-hub areas and by 0.05 percent in non–trade hub areas. These findings are consistent with our main results and confirm the robustness of our analysis.

As a third robustness check, we exclude the United States and Canada from the analysis. Measuring sea distance for these two countries is particularly sensitive to port selection, as both border the Atlantic and Pacific Oceans. In our baseline analysis, we assume that countries located near the Atlantic Ocean trade with the United States and Canada via the ports of New York and Montreal, while countries near the Pacific Ocean trade via the ports of Los Angeles and Vancouver. To test the robustness of this assumption, we exclude the

Table 11. The Second Stage Estimated Effects of Exports on Lights in Trade Hub Area & Non-Trade Hub Area: Results Within a 50 km Radius Buffer Area After Removing the Central 3 km Radius Around Trade Hubs

	(1)	(2)	(3)	(4)	(5)	(6)
Area	Trade Hub Area			Non-Trade Hub Area		
A. Second-Stage Estimates						
Dependent Variable	ln Lights			ln Lights		
ln Exports	0.300***	0.302***	0.305***	0.0585	0.0630	0.0558
	(0.0672)	(0.0660)	(0.0670)	(0.0774)	(0.0772)	(0.0809)
B. Reduced Form Estimates						
ln Predicted Exports	0.302***	0.303***	0.301***	0.0588	0.0633	0.0547
	(0.0816)	(0.0808)	(0.0809)	(0.0829)	(0.0830)	(0.0844)
R-Squared	0.987	0.987	0.987	0.998	0.998	0.998
C. Specifications						
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
National Population		Yes	Yes		Yes	Yes
Local Weather			Yes			Yes
Countries	101	101	101	101	101	101
N	505	505	505	505	505	505

Notes: The control variables include country-level population and local weather for trade hub and non-trade hub areas. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

United States and Canada entirely and rerun the regressions. The results, presented in Supplementary Information C, remain consistent with our main findings in both magnitude and statistical significance.

Finally, to address the concern that our results may simply reflect regional differences in economic growth combined with variations in sea and air distances across continents, we conduct a fourth robustness check by excluding countries from one region at a time and re-estimating the model. For example, in one case, we exclude all countries in Central and South America; in another, we exclude all Sub-Saharan African countries. If the estimated coefficients from these exercises are similar to those in the main results, it suggests that re-

Table 12. The Second Stage Estimated Effects of Exports on Population Ratio in Trade Hub & Non-Trade Hub Areas: Results Within a 50 km Buffer Area, Excluding the Central 3 km Radus Around Trade Hubs

	(1)	(2)	(3)	(4)	(5)	(6)
Area	Trade Hub Area			Non-Trade Hub Area		
A. Second-Stage Estimates						
Dependent Variable	Population Ratio			Population Ratio		
ln Exports	0.00698	0.00657	0.00906	-0.00698	-0.00657	-0.00906
	(0.0127)	(0.0124)	(0.0144)	(0.0127)	(0.0124)	(0.0144)
B. Reduced Form Estimates						
ln Predicted Exports	0.00703	0.00660	0.00891	-0.00703	-0.00660	-0.00891
	(0.0128)	(0.0125)	(0.0143)	(0.0128)	(0.0125)	(0.0143)
R-Squared	0.993	0.993	0.993	0.993	0.993	0.993
C. Specifications						
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
National Population		Yes	Yes		Yes	Yes
Local Weather			Yes			Yes
Countries	101	101	101	101	101	101
N	505	505	505	505	505	505

Notes: The control variables include country-level population and local weather for trade hub and non-trade hub areas. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

gional differences are not driving our findings. Figures D1–D7 present the results of these regressions, which are consistent with the main findings reported in Tables 7, 8, and 9, reinforcing the robustness of our conclusions.

Table 13. The Second Stage Estimated Effects of Exports on Lights per Capita in Trade Hub & Non-Trade Hub Areas: Results Within a 50 km Radius Buffer Area After Removing the Central 3 km Radius Around Trade Hubs

	(1)	(2)	(3)	(4)	(5)	(6)
Area	Trade Hub Area			Non-Trade Hub Area		
A. Second-Stage Estimates						
Dependent Variable	ln Lights per Capita			ln Lights per Capita		
ln Exports	0.168	0.199***	0.196***	0.0161	0.0490	0.0439
	(0.106)	(0.0662)	(0.0682)	(0.121)	(0.0817)	(0.0857)
B. Reduced Form Estimates						
ln Predicted Exports	0.169	0.200**	0.193**	0.0162	0.0633	0.0430
	(0.111)	(0.0769)	(0.0770)	(0.123)	(0.0830)	(0.0882)
R-Squared	0.977	0.986	0.986	0.988	0.998	0.994
C. Specifications						
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
National Population		Yes	Yes		Yes	Yes
Local Weather			Yes			Yes
Countries	101	101	101	101	101	101
N	505	505	505	505	505	505

Notes: The control variables include country-level population and local weather for trade hub and non-trade hub areas. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

4 Discussion

This study examines the effects of exports on subnational economic development and rural–urban inequality using nightlight intensity data. Trade-hub areas are defined as the union of areas within either a 50 km or a 30 km radius of each of the three major ports and three international airports in a country. All other areas are classified as non–trade-hub areas.

Using a 50 km radius definition, we find that a 1% increase in exports raises nightlight intensity by 0.31% in trade-hub areas and by 0.06% in non–trade-hub areas, resulting in a 0.25% increase in the rural–urban difference in nightlight intensity. Similarly, a 1% increase in exports raises nightlight intensity per capita by 0.19% in trade-hub areas and by 0.04% in non–trade-hub areas. When we exclude 3 km circles around ports and airports to remove direct light emissions from these facilities, the results remain robust. Furthermore, when defining trade-hub areas using a 30 km radius instead of 50 km, we find that a 1% increase in exports raises nightlight intensity by 0.32% in trade-hub areas and by 0.06% in non–trade-hub areas.

Between 1997 and 2018 (just before the COVID-19 pandemic), exports grew by an average of 380% across the countries included in our analysis. Based on our estimation results, nightlight intensity per capita—a proxy for economic activity—increased by 72.2% in trade-hub areas and by 15.2% in non–trade-hub areas. Consequently, the gap in nightlight intensity per capita between trade-hub and non–trade-hub areas widened by 57%, which is substantial. As shown in Table 1, a 1% increase in nightlight intensity is associated with a 0.26–0.30% increase in GDP. Thus, a 57% increase in the difference in nightlight intensity per capita can be interpreted as a 15–17% increase in the income gap.

According to our data, in the year 2000, an average of 45 percent of each country's population resided in trade hub areas. Meanwhile, Table 7 shows that the migration from non-trade hub area to trade hub area is small. Thus, our estimates suggest that the 56 percent of the population who originally lived in non-trade hub areas did not benefit from the export growth as much as the 45 percent who were already living in trade hub areas.

Our results are consistent with the findings of Storeygard (2016) but stand in sharp contrast to those of Hirte et al. (2020). Storeygard (2016) found that, with rising oil prices, the income of cities located closer to a major port increased by 7% relative to cities located 500 km farther from the port within the same country.

By contrast, Hirte et al. (2020) find that international trade does not have an unconditional causal effect on inter-regional income inequality. Several factors may explain why our estimation results differ. First, while Hirte et al. (2020) use a Bartik-type instrument, we employ a dynamic gravity equation in which variation is driven by exogenous changes in transportation costs and technology. Different instrumental variables can yield different estimates of local average treatment effects. Second, and more importantly, Hirte et al. (2020) focus on inter-regional inequality and measure it using the Gini index. The Gini index captures income disparities not only between urban and rural areas but also among rural areas themselves. In contrast, our study specifically compares urban (trade hub) areas with rural (non–trade hub) areas. Taken together, the results of Hirte et al. (2020) and our study suggest that globalization does not necessarily increase inequality among all regions, but it does widen the gap between trade hub and non–trade hub areas.

As discussed in the introduction, in the 2024 U.S. presidential election, the Republican candidate—who criticized globalization—dominated inland regions, while the Democratic candidate—who supported globalization—secured the majority of votes in areas close to the coasts (The Wall Street Journal, 2024; National Association of Counties, 2024). This pattern is consistent with our findings and highlights the importance of focusing on trade hub versus non–trade hub inequality, rather than inter-regional inequality.

A natural question is why people do not move from non–trade hub areas to trade hub areas when a country experiences globalization. Our analysis indicates that such migration does occur, but its scale is relatively small. In practice, several obstacles hinder movement from non–trade hub areas to trade hub areas, including information asymmetry, higher rents in trade hub areas due to limited housing supply, and skill mismatches. This suggests that it is important to study the barriers to migration in the context of globalization, as this has significant policy implications.

5 Conclusion

We find that a 1 percent increase in exports raises nightlight intensity by 0.31 percent in trade-hub areas and by 0.06 percent in non–trade hub areas, widening the rural–urban income gap by an estimated 15–17 percent between 1997 and 2020. Economic activity within countries is highly uneven, with nightlight data revealing that most activity is concentrated along coastlines or in inland urban centers, forming distinct trade-hub and non–trade hub regions. While exports substantially boost economic performance in trade-hub regions, their effects in non–trade hub areas are much more limited. As countries pursue export-led growth strategies, these findings underscore the importance of inclusive policies that extend the benefits of trade beyond urban centers, ensuring prosperity is shared across the entire national landscape.

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Supplemental Information A

Figure A1: Nighttime Lights Map of South America in 2010

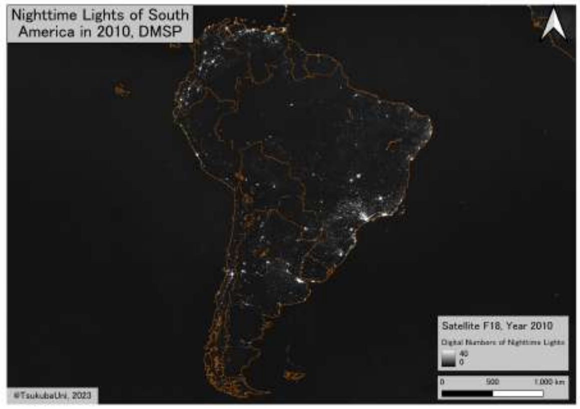

Figure A2: Nighttime Lights Map of Australia in 2010

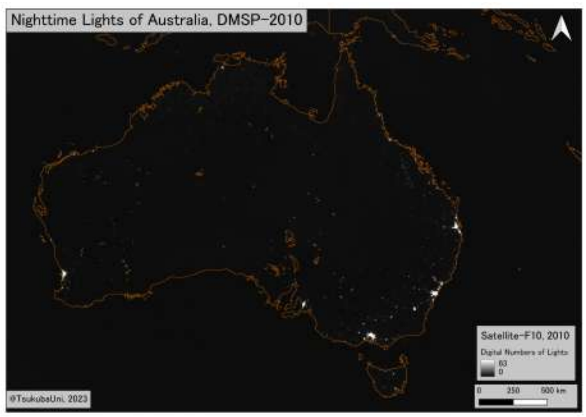

Figure A3: Nighttime Lights Map of Sub-Saharan Africa in 2010

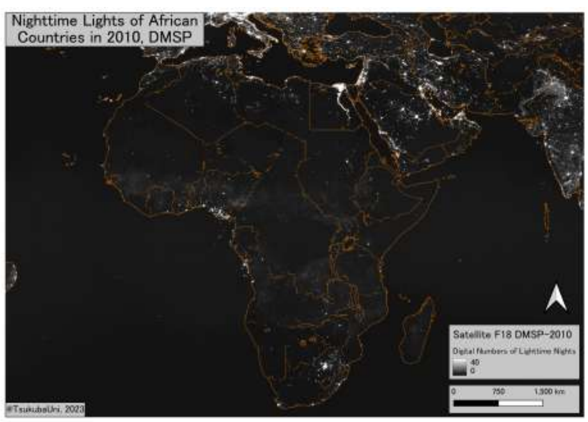

Figure A4: Nighttime Lights Map of South East Asia in 2010

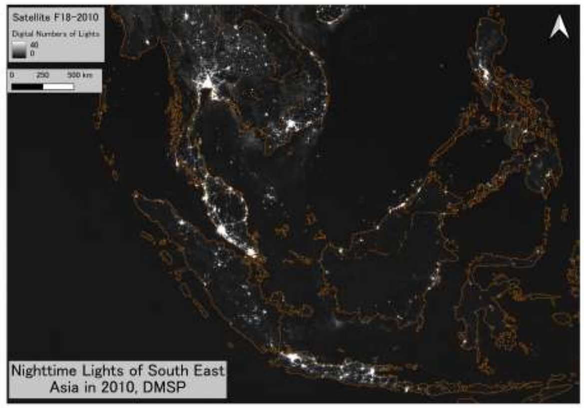

Figure A5: Nighttime Lights Map of East Asia in 2010

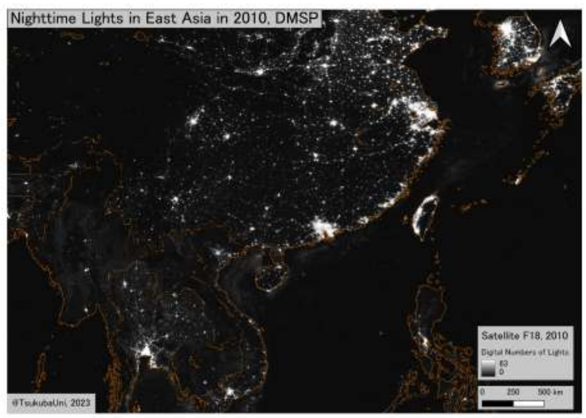

Figure A6: Nighttime Lights Map of Europe in 2010

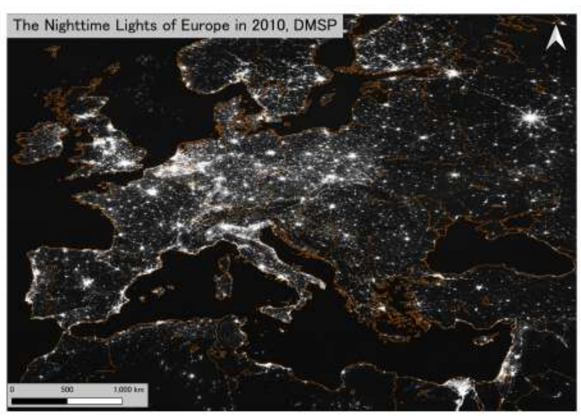

Table A1: List of Countries included in the Study

1	Albania	27	Dominican Rep	53	Ivory Coast	79	Philliphines
2	Algeria	28	Ecuador	54	Jamaica	80	Poland
3	Angola	29	Egypt	55	Japan	81	Portugal
4	Argentinia	30	El Salvador	56	Jordan	82	Romania
5	Australia	31	Estonia	57	Kenya	83	Russian Federation
6	Bahamas	32	Fiji	58	Korea Rep	84	Senegal
7	Bangladesh	33	Finland	59	Latvia	85	Sierra Leone
8	Belgium	34	France	60	Lebanon	86	South Africa
9	Belize	35	Gambia	61	Liberia	87	Spain
10	Benin	36	Georgia	62	Lithuania	88	Sri Lanka
11	Brazil	37	Germany	63	Madagascar	89	Sweden
12	Brunei	38	Ghana	64	Malaysia	90	Syria
13	Bulgaria	39	Greece	65	Mauritania	91	Tanzania
14	Cambodia	40	Guatemala	66	Mexico	92	Thailand
15	Cameroon	41	Guinea	67	Morocco	93	Togo
16	Canada	42	Guinea Bassau	68	Mozambique	94	Trinidad & Tobago
17	Chile	43	Guyana	69	Namibia	95	Tunisia
18	China	44	Haiti	70	Netherlands	96	Turkey
19	Colombia	45	Honduras	71	New Zealand	97	Ukraine
20	Congo	46	Iceland	72	Nicaragua	98	United Kingdom
21	Congo DRC	47	India	73	Nigeria	99	United States
22	Costa Rica	48	Indonesia	74	Norway	100	Uruguay
23	Croatia	49	Iran	75	Papu New Guinea	101	Vietnam
24	Cuba	50	Iraq	76	Pakistan		
25	Cyprus	51	Ireland	77	Panama		
26	Denmark	52	Italy	78	Peru		

Table A2. Second Stage Estimated Effects of Exports on the Difference in ln Lights Between Trade Hub Area and Non-Trade Hub Area

	(1)	(2)	(3)
A. Second-Stage Estimates			
Dependent Variable	Difference of ln Lights of Trade Hub & Non-Trade Hub Areas		
Ln Exports	0.236***	0.233***	0.250***
	(0.0511)	(0.0533)	(0.0552)
B. Reduced Form Estimates			
Ln Predicted Exports	0.238***	0.234***	0.245***
	(0.0380)	(0.0403)	(0.0402)
R-Squared	0.998	0.998	0.998
C. Specifications			
Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
National Population		Yes	Yes
National Weather			Yes
Countries	101	101	101
N	505	505	505

Notes: The control variables include country-level population and weather. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

Table A3. Second Stage Estimated Effects of Exports on the Difference in Population Between Trade Hub Area and Non-Trade Hub Area

	(1)	(2)	(3)
A. Second-Stage Estimates			
Dependent Variable	Difference of ln Population of Trade Hub & Non-trade Hub Areas		
Ln Exports	0.0892	0.0866	0.0974
	(0.0714)	(0.0721)	(0.0802)
B. Reduced Form Estimates			
Ln Predicted Exports	0.0897	0.0870	0.0957
	(0.0727)	(0.0734)	(0.0806)
R-Squared	0.995	0.995	0.995
C. Specification			
Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
National Population		Yes	Yes
National Weather			Yes
Countries	101	101	101
N	505	505	505

Notes: The control variable weather. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

Table A4. Second Stage Estimated Effects of Exports on the Difference in Lights per Capita Between Trade Hub Area and Non-Trade Hub Area

	(1)	(2)	(3)
A. Second-Stage Estimates			
Dependent Variable	Difference of ln Lights per Capita of Trade Hub & Non-trade Hub Areas		
Ln Exports	0.147*	0.146*	0.152*
	(0.0757)	(0.0764)	(0.0854)
B. Reduced Form Estimates			
Ln Predicted Exports	0.148**	0.147**	0.150*
	(0.0715)	(0.0718)	(0.0787)
R-Squared	0.984	0.985	0.985
C. Specifications			
Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
National Population		Yes	Yes
National Weather			Yes
Countries	101	101	101
N	505	505	505

Notes: The control variables include country-level population and weather. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

Table A5. Second Stage Estimated Effects of Exports on the Difference in Lights Between Trade Hub & Non-Trade Hub Areas: Results Within a 50 km Radius Buffer Area After Removing the Central 3 km Radius Around Trade Hubs

	(1)	(2)	(3)
A. Second-Stage Estimates			
Dependent Variable	Difference of ln Lights of Trade Hub & Non-Trade Hub Areas		
Ln Exports	0.242***	0.239***	0.255***
	(0.0508)	(0.0530)	(0.0551)
B. Reduced Form Estimates			
Ln Predicted Exports	0.243***	0.240***	0.251***
	(0.0382)	(0.0406)	(0.0406)
R-Squared	0.998	0.998	0.998
C. Specifications			
Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
National Population		Yes	Yes
Local Weather			Yes
Countries	101	101	101
N	505	505	505

Notes: The control variables include country-level population and weather. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

Table A6. Second Stage Estimated Effects of Exports on the Difference in Population Between Trade Hub & Non-Trade Hub Area: Results Within a 50 km Radius Buffer Area After Removing the Central 3 km Radius Around Trade Hubs

	(1)	(2)	(3)
A. Second-Stage Estimates			
Dependent Variable	Difference of ln Population of Trade Hub & Non-trade Hub Areas		
Ln Exports	0.0901	0.0885	0.103
	(0.0675)	(0.0674)	(0.0740)
B. Reduced Form Estimates			
Ln Predicted Exports	0.0906	0.0889	0.101
	(0.0686)	(0.0685)	(0.0741)
R-Squared	0.994	0.994	0.994
C. Specification			
Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
National Population		Yes	Yes
Local Weather			Yes
Countries	101	101	101
N	505	505	505

Notes: The control variable weather. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

Table A7. Second Stage Estimated Effects of Exports on the Difference in Lights per Capita Between Trade Hub & Non-Trade Hub Area: Results Within a 50 km Radius Buffer Area After Removing the Central 3 km Radius Around Trade Hubs

	(1)	(2)	(3)
A. Second-Stage Estimates			
Dependent Variable	Difference of ln Lights per Capita of Trade Hub & Non-trade Hub Areas		
Ln Exports	0.152**	0.150**	0.152*
	(0.0717)	(0.0715)	(0.0783)
B. Reduced Form Estimates			
Ln Predicted Exports	0.153**	0.151**	0.150**
	(0.0675)	(0.0669)	(0.0720)
R-Squared	0.987	0.987	0.987
C. Specifications			
Country FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
National Population		Yes	Yes
Local Weather			Yes
Countries	101	101	101
N	505	505	505

Notes: The control variables include country-level population and weather. Weather includes precipitation and temperature. Clustered robust standard errors are reported in parentheses, assuming that the error terms are correlated within each country. *** indicates significance at the 1% level, ** at the 5% level, and * at the 10% level.

Table A8. Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Summed Lights	272455	5034.109	5413.211	921.309	38619.094
Population (sum)	272455	450682.8	804270.19	135.541	5885810
Lights Per Capita	272455	0.05	0.177	0.003	15.447
Electricity Access	272455	0.277	0.447	0	1
Radio Ownership	272455	0.614	0.487	0	1
Television Ownership	272455	0.233	0.423	0	1
Motorcycle Ownership	272455	0.086	0.28	0	1
Bicycle Ownership	272455	0.275	0.447	0	1
Good Floor Materials Ownership	272455	0.426	0.495	0	1
Improved Cooking Fuel	272455	0.098	0.298	0	1
Improved Sanitation Facility	272455	0.109	0.311	0	1
Improved Drinking Water Source	272455	0.649	0.477	0	1
Household Head Age	272455	44.497	15.917	9	94
Household Head Gender	272455	0.711	0.454	0	1
Household Head Years of Schooling	272455	5.064	4.861	0	25
Household Size	272455	4.77	2.785	1	53

Note: Lights are calculated for each cluster using a 10-kilometre radius.

Table A9. Countries Included in the Sample and DHS Survey Waves Used in the Analysis

#	Country	Wave 1	Wave 2	Wave 3	Wave 4
1	Benin	2001	2011		
2	Burkina Faso	2003	2010		
3	Burundi	2010	2012		
4	Cameroon	2004	2011	2018	
5	Ethiopia	1997	2000	2005	
6	Ghana	2003	2008	2014	
7	Guinea	2005	2012		
8	Kenya	2003	2008	2014	
9	Lesotho	2004	2009	2014	
10	Malawi	2000	2004	2010	2015-16
11	Mali	2000	2006	2012-13	
12	Rwanda	2005	2010	2014-15	
13	Namibia	2000	2006	2013	
14	Sierra Leone	2008	2013		
15	Tanzania	2007	2010	2011	2015
16	Uganda	2000	2006	2011	
17	Zambia	2007	2013-14		
18	Zimbabwe	1999	2005	2010	2015

Notes: The sample includes Sub-Saharan African countries with at least two geocoded DHS survey waves that can be matched to DMSP nighttime lights data.